

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO5 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
GRAND TOTAL:			20 Marks

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

- (a) Interpret the scenario and identify the type of noise present.**
- (b) Explain how this degradation affects inspection accuracy.**
- (c) Recommend an appropriate enhancement technique to remove the interference.**
- (d) Justify why your selected approach is more suitable than spatial-domain methods.**
- (e) Explain your answer in a logical sequence.**

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

- (a) Interpret the scenario and explain the challenge involved.**
- (b) Identify the importance of preserving edges while removing noise.**
- (c) Recommend a suitable filtering technique.**
- (d) Justify why your selected filter preserves important image details.**
- (e) Present your explanation clearly and systematically.**

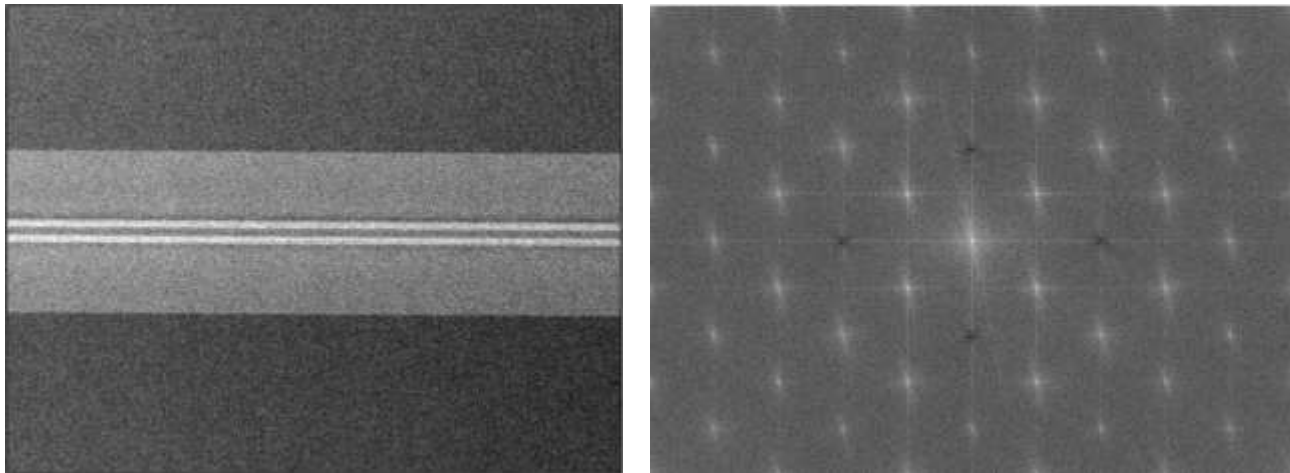
Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)		Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

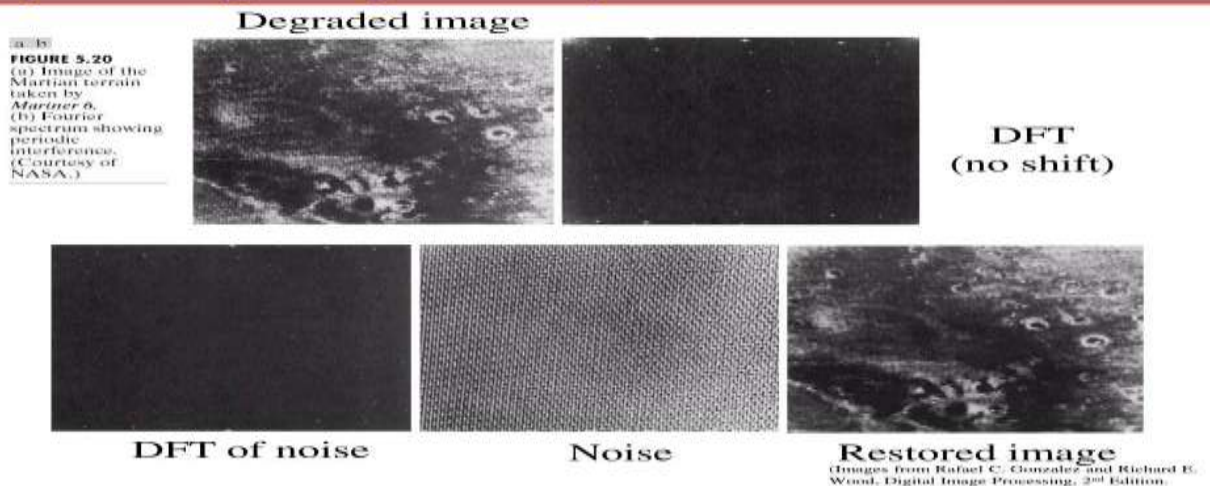
Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Scenario 1: Manufacturing Quality Inspection (10 Marks)

(a) Interpret the scenario and identify the type of noise present.



Example: Image Degraded by Periodic Noise



An automated quality inspection system captures images of metal components to detect defects such as scratches, cracks, and dents. Due to electrical interference during image acquisition, repetitive horizontal stripes appear across the images.

Type of Noise:

- **Periodic Noise (Sinusoidal Noise)**
- Caused by electrical interference from power supplies, motors, or electromagnetic signals.
- Appears as regularly repeating horizontal or vertical stripes.

(b) Explain how this degradation affects inspection accuracy.

Periodic noise reduces the quality of the captured image by:

- Hiding small defects such as cracks and scratches.
- Reducing image contrast.
- Producing false edges that may be mistaken for defects.
- Lowering the accuracy of automatic inspection systems.
- Increasing false positives and false negatives during defect detection.

Result: The manufacturing system becomes less reliable and may classify defective products as good or vice versa.

(c) Recommend an appropriate enhancement technique to remove the interference.

Recommended Technique:

Frequency Domain Filtering using a Notch Reject Filter

Working:

1. Convert the image into the frequency domain using the **Fourier Transform (FFT)**.
2. Locate the bright peaks corresponding to periodic noise.
3. Apply a **Notch Reject Filter** to remove only those unwanted frequencies.
4. Perform the **Inverse FFT** to reconstruct the clean image.

(d) Justify why your selected approach is more suitable than spatial-domain methods.

Frequency Domain (Notch Filter) Spatial Domain Filters

Removes only periodic frequencies Removes pixels locally

Preserves image details Blurs important features

Very effective for stripe noise Poor for periodic interference

High defect detection accuracy May remove useful information

Justification:

Periodic noise has a specific frequency pattern. The Fourier Transform clearly separates these frequencies, allowing only the unwanted components to be removed without affecting the actual image.

Spatial filters (Mean, Median, Gaussian) cannot distinguish periodic interference from image details, so they often blur both the noise and the defects.

(e) Explain your answer in a logical sequence.

Flow of Enhancement

Capture Metal Image



Horizontal Stripe Noise Added



Apply Fourier Transform (FFT)



Detect Noise Frequencies



Apply Notch Reject Filter



Inverse FFT



Enhanced Image



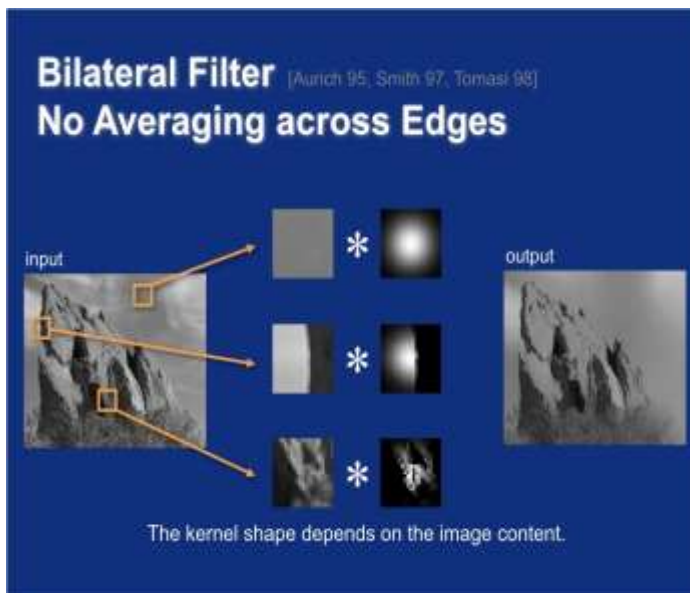
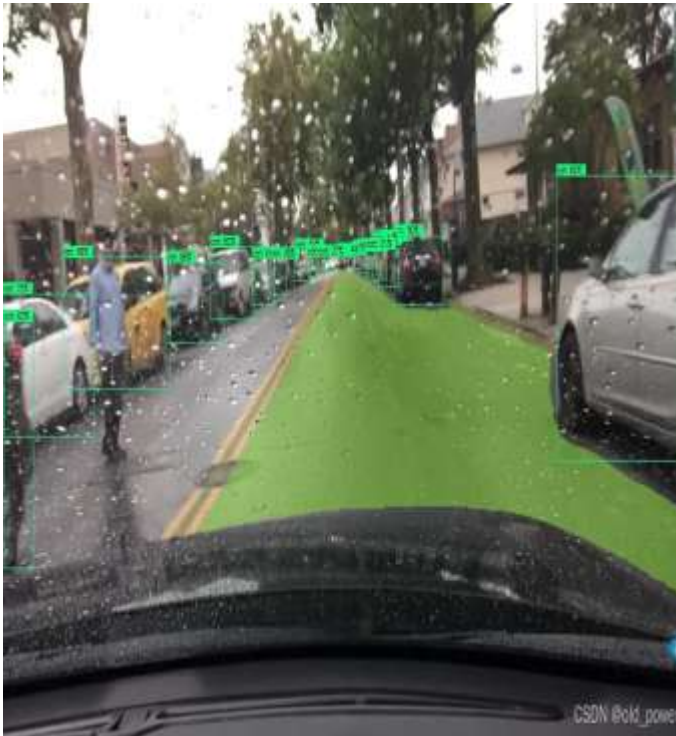
Accurate Defect Detection

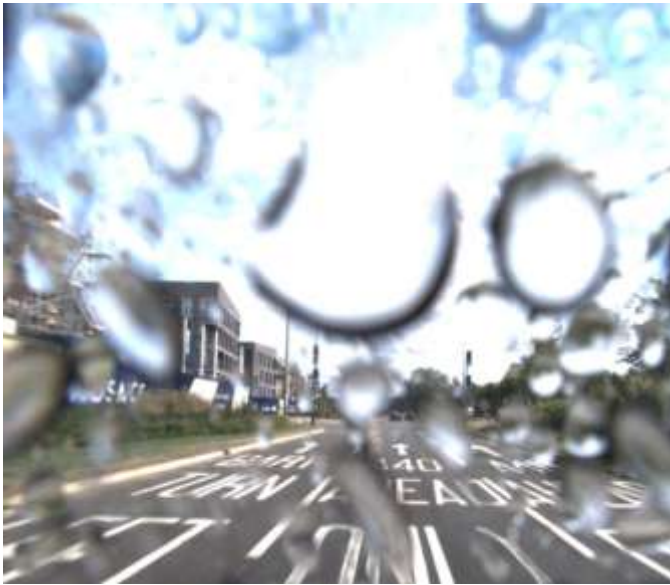
Conclusion

Using a Notch Reject Filter in the frequency domain effectively removes periodic stripe noise while preserving important defect information, making it the best choice for industrial quality inspection.

Scenario 2: Autonomous Vehicle Navigation (10 Marks)

(a) Interpret the scenario and explain the challenge involved.





A self-driving car captures road images during heavy rain. Rain introduces noise and reduces image clarity, making it difficult to identify lane markings, road boundaries, and nearby objects.

Challenge

The enhancement technique must:

- Remove rain-induced noise.
- Preserve lane markings.
- Preserve road edges.
- Maintain object boundaries.
- Support accurate autonomous navigation.

(b) Identify the importance of preserving edges while removing noise.

Edges represent important features such as:

- Lane markings
- Road boundaries
- Vehicles
- Pedestrians
- Traffic signs

If edges become blurred:

- Lane detection becomes inaccurate.
- Road boundaries disappear.
- Object recognition performance decreases.
- Navigation decisions become unsafe.

Therefore, edge preservation is essential for safe autonomous driving.

(c) Recommend a suitable filtering technique.

Recommended Filter

Bilateral Filter

The Bilateral Filter performs smoothing while preserving image edges.

It considers:

- Spatial distance between pixels.
- Intensity difference between neighboring pixels.

Thus, only similar pixels are averaged together.

(d) Justify why your selected filter preserves important image details.

Bilateral Filter	Mean/Gaussian Filter
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Removes noise	Removes noise
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Preserves edges	Blurs edges
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Maintains lane markings	Weakens lane markings
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Better object boundaries	Loss of detail
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Ideal for autonomous driving	Not suitable for edge-critical tasks
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Justification

Unlike ordinary smoothing filters, the Bilateral Filter does not average pixels across strong edges.

As a result:

- Lane markings remain sharp.
- Road boundaries stay visible.
- Objects maintain their shapes.
- Navigation accuracy improves.

(e) Present your explanation clearly and systematically.

Processing Flow

Capture Road Image



Rain Noise Present



Apply Bilateral Filter



Noise Reduced



Edges Preserved



Lane Detection



Safe Vehicle Navigation

Conclusion

The **Bilateral Filter** is the most suitable enhancement technique because it removes rain noise while preserving critical edges such as lane markings and road boundaries, ensuring reliable autonomous vehicle navigation.

Summary

Scenario	Noise Type	Best Enhancement Technique	Reason
Manufacturing Quality Inspection	Periodic (Stripe) Noise	Notch Reject Filter (Frequency Domain)	Removes periodic interference without affecting defects
Autonomous Vehicle Navigation	Rain Noise	Bilateral Filter	Removes noise while preserving edges and lane markings